Problem Set 3 · Preferences and rationality

Microeconomics (040184) · Group 1 · WS 2026/27 · **Solutions**

All problems are about preferences (L06). Where a problem uses an idea we have not covered in class, a blue box explains it before you need it. Problem 4 (d) is optional, but its content can come up in the midterm and final.

Problem 1 · Preferences and indifference curves

From the lecture (L06).

- $A \gtrsim C$: A is at least as good as C; $A \succ C$: strictly better; $A \sim C$: equally good.
- **Complete**: any two options can be compared (one is liked more, or both equally). **Transitive**: if $A \gtrsim C$ and $C \gtrsim D$, then $A \gtrsim D$.
- **More is better** (strong monotonicity): at least as much of everything, and strictly more of something, is strictly better.
- An **indifference curve** joins bundles the person likes equally. Curves further out are better.

(a) A friend talks about three lunch dishes: pasta, salad and soup.

1. "Pasta or salad? I like them exactly the same."
2. "Pasta or soup? I can't compare them at all; I just can't say."
3. "Salad is strictly better than soup."

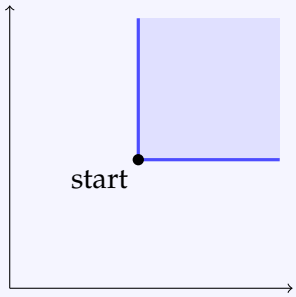
Which statements show that the friend's preferences are **not complete**?

- ☐ 1 only
- ☐ 2 only
- ☐ 1 and 2
- ☐ 2 and 3
- ☐ 1, 2 and 3

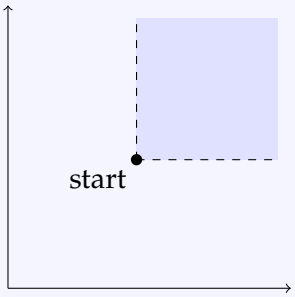
More is better: what exactly does it mean? Three versions.
Do you need this box for the next question? Only version 1, which is "more is better" as we used it in class. Versions 2 and 3 show what it does *not* say. *Do we ever need to assume local non-satiation?* No: it follows from version 1 (see below).
Bundles are (cinema tickets, euros left). Start from the bundle $(7, 15)$.

1. **More is better, strong** (strong monotonicity). At least as much of every good, and strictly more of at least one, is strictly better. Example: $(8, 15) \succ (7, 15)$: one more ticket, the same euros. **This is "more is better" as in our lectures and on this problem set.**
2. **More is better, weak** (weak monotonicity). Strictly more of *every* good is strictly better. Example: $(8, 16) \succ (7, 15)$. It says nothing about $(8, 15)$. Someone with perfect complements (sausages and buns for hot dogs) satisfies the weak version but not the strong one: 3 sausages and 2 buns are no better than 2 and 2.
3. **A weaker cousin: local non-satiation.** However close you look around a bundle, there is a strictly better bundle nearby. It does *not* say in which direction. Example: coffees and minutes of noisy commute. More of everything is *not* better (more noise

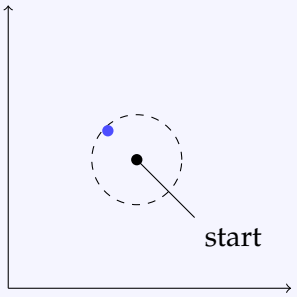
hurts), so more is better fails. But a little more coffee, or a little less noise, is always better, so local non-satiation holds.



Strong: the shaded area *and* its edges have to be better.



Weak: only the inside has to be better.

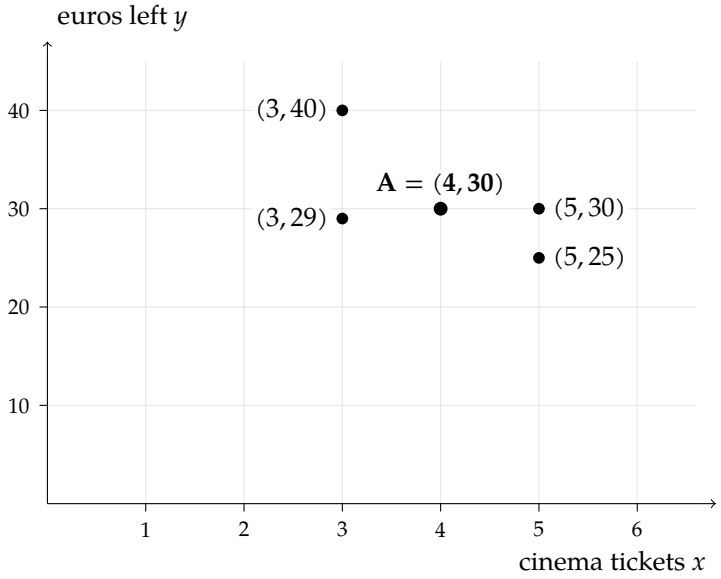


Non-satiation: something better is nearby.

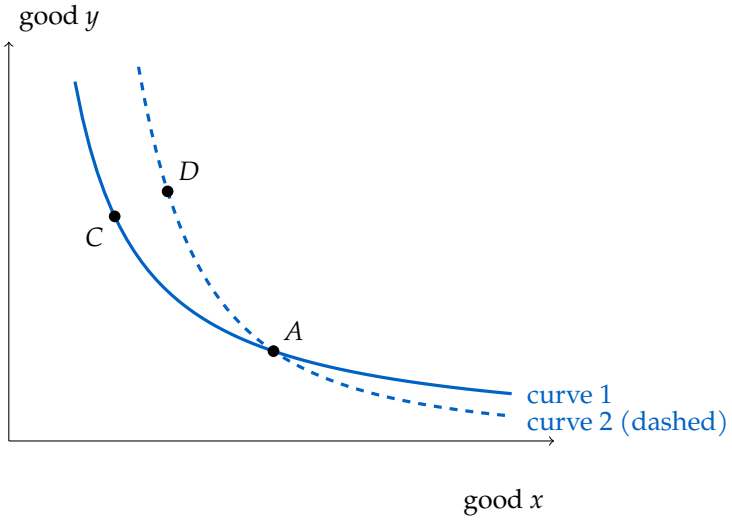
How they relate. Strong ⇒ weak ⇒ local non-satiation. None of the arrows can be reversed. So with the strong version we get the other two for free: **we never need to assume local non-satiation separately.** It matters only because it is the *weakest* version that still makes "money left over, so strictly better" work.

(b) Bundles are (cinema tickets, euros left). Your bundle is $A = (4, 30)$. Using only more is better, which of these bundles is better than A?

- ☐ $(3, 40)$
- ☐ $(5, 25)$
- ☐ $(4, 30)$
- ☐ $(5, 30)$
- ☐ $(3, 29)$



(c) A friend draws two of her indifference curves, and they cross at bundle A. C is on curve 1, D is on curve 2, and D has more of both goods than C.



1. $C \sim A$ and $A \sim D$.
2. By transitivity, $C \sim D$.
3. This drawing can come from preferences that follow our properties.

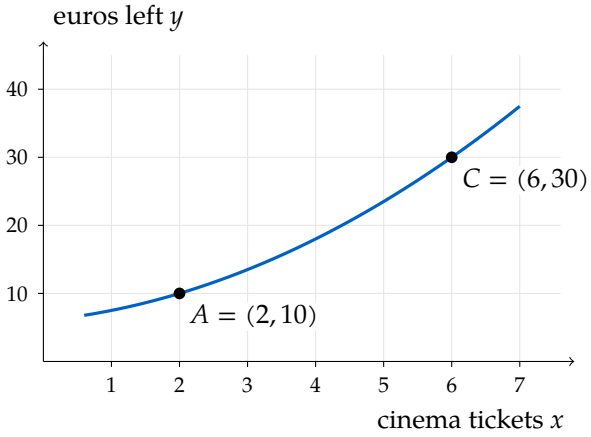
Which statements are correct?

- ☐ 1 only
- ☐ 2 only
- ☐ 1 and 2
- ☐ 1 and 3
- ☐ 1, 2 and 3

Which properties rule out this drawing?

- ☐ completeness alone
- ☐ transitivity alone
- ☐ more is better alone
- ☐ transitivity and more is better
- ☐ none: such a drawing is possible

(d) A friend draws one of her indifference curves for (cinema tickets, euros left). It slopes **upward**: it passes through $A = (2, 10)$ and $C = (6, 30)$.



(i) Compared with A, bundle C has

- ☐ less of both goods
- ☐ more of both goods
- ☐ more of one good and less of the other

(ii) Using only more is better, C is

- ☐ equally good as A
- ☐ worse than A
- ☐ strictly better than A

(iii) Because A and C lie on the same indifference curve, the friend says C and A are

- ☐ equally good
- ☐ A strictly better
- ☐ C strictly better

(iv) So an upward-sloping indifference curve breaks which property (when both are goods)?

- ☐ completeness
- ☐ transitivity
- ☐ more is better
- ☐ convexity

Further reading (*optional*)*:* Our textbook covers this in R. H. Frank, *Microeconomics and Behavior*, 9th ed., ch. 3: preferences and their assumptions, indifference curves and the marginal rate of substitution; see pp. 63–69.

Problem 2 · Keep it or give it

From the lecture (L06).

- A split is *affordable* on a day if it costs at most that day's money. The *budget line* $p_x\, x + y = B$ contains the splits that use all the money.
- Preferences are **rational** if they are **complete** (any two options can be compared: one is liked more, or both equally) and **transitive** (if $A \gtrsim C$ and $C \gtrsim D$, then $A \gtrsim D$).
- **More is better**: an option with at least as much of everything, and strictly more of something, is strictly better.

New: learning preferences from choices (revealed preference).

- If a person could afford option C but chose A, the choice *reveals* that A is **at least as good** as C: $A \gtrsim C$ (*weakly* preferred).
- If, in addition, C would leave money unused while A uses all the money, and more is better for the person, then A is **strictly better** than C: $A \succ C$. (With the unused money one could have had more than C, which is strictly better than C; and A was chosen over that too.)

The story. Aman splits money between **himself** and **Tony**. Let

$$x = \text{euros Tony gets}, \qquad y = \text{euros Aman keeps}.$$

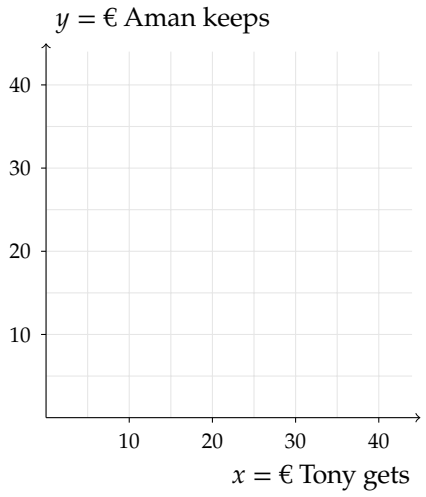
Keeping €1 costs €1. Giving Tony €1 costs p_x, and this price changes from day to day. Any money Aman does not use is **burned**: nobody gets it.

Day 1. Aman has **€40**. Whatever he sends is halved (he sends €2, Tony gets €1), so $p_x = 2$.
Day 2. Aman has **€30**. Whatever he sends arrives in full (he sends €1, Tony gets €1), so $p_x = 1$.

Assumption. For Aman, **more is better**: he likes more euros for himself *and* more euros for Tony. So an option that gives at least as much to both, and strictly more to one, is strictly better for him. In particular, he never wants to burn money.

Part A · Aman

(a) Draw the two budget lines in the graph below, with their end points.



(b) Look at two splits, $a = (15, 10)$ and $a' = (12, 8)$. Mark both in the graph. Aman prefers a to a'. Which property of his preferences makes this reasonable?

- ☐ transitivity
- ☐ more is better
- ☐ completeness

(c) In order to learn about Aman's preferences from his choices, he must be able to say, for any two options, whether he likes one more or likes them equally. Which property are we assuming about Aman's preferences?

- ☐ transitivity
- ☐ more is better
- ☐ completeness

(d) On day 1, Aman chooses $a = (15, 10)$, which uses all his €40.

(i) Is $a' = (12, 8)$ affordable on day 1?

[yes / no]

(ii) How much money would a' leave unused (burned) on day 1? € _____

(iii) Given that more is better for Aman, what does his choice on day 1 tell us?

- ☐ a is only weakly better than a'
- ☐ a is strictly better than a'
- ☐ nothing
- ☐ a' is strictly better than a

(e) On day 2, Aman chooses $b = (5, 25)$, which uses all his €30. Mark b in the graph.

(i) Could Aman afford a (his day-1 choice) on day 2?

[yes / no]

(ii) How much money would a leave unused (burned) on day 2? € _____

(iii) Given that more is better for Aman, what does his choice on day 2 tell us?

- ☐ b is strictly better than a
- ☐ nothing
- ☐ a is strictly better than b
- ☐ b is only weakly better than a

(f) Go back to day 1. Could Aman afford b (his day-2 choice) on day 1?

[yes / no]

How much money would b leave unused (burned) on day 1? € _____
Given that more is better for Aman, what does this tell us about his preferences?

- ☐ b is strictly better than a
- ☐ nothing
- ☐ a is only weakly better than b
- ☐ a is strictly better than b

(g) Put day 1 and day 2 together. Given that more is better for Aman, is Aman's altruism rational?

[yes / no]

If not, which property of rational preferences does it fail?

- ☐ completeness
- ☐ convexity
- ☐ transitivity
- ☐ it does not fail any property

Now drop *every* form of "more is better": assume nothing about it (Aman might even be happy to burn money). Could Aman's two choices come from rational (complete and transitive) preferences?

[yes / no]

Part B · Andrey

Andrey faces the same two days and more is better for him too. He chooses $(5, 30)$ on day 1 and $(8, 22)$ on day 2.

(h) Is Andrey's altruism rational?

[yes / no]

Part C · How far is Aman from rational?

Go back to Aman (Part A). On day 1 he chose $a = (15, 10)$ with €40; on day 2 he chose $b = (5, 25)$ with €30, and these choices could not come from rational preferences for which more is better. Now we ask: *how much money would we have to take away from Aman, so that his choices stop contradicting each other?* Taking money away keeps the price the same, so it shifts a budget line **inward, parallel** to itself.

(i) **Shift the day-1 line only.** On day 1 the price stays $p_x = 2$, but Aman's money is B_1 instead of €40.

(i) What does $b = (5, 25)$ cost at day-1 prices? € _____

(ii) For which amounts B_1 is b *no longer* affordable on day 1? $B_1 <$ € _____

(iii) If b is not affordable on day 1, does day 1 still tell us that a is strictly better than b?

[yes / no]

(iv) Are Aman's choices then rational?

[yes / no]

(v) As a fraction of the original €40: below which fraction of the money is b no longer affordable on day 1? _____ (write it as a decimal).

(j) **Shift the day-2 line only** (keep day 1 at €40). On day 2 the price stays $p_x = 1$, but Aman's money is B_2 instead of €30.

(i) What does $a = (15, 10)$ cost at day-2 prices? € _____

(ii) For which amounts B_2 is a no longer affordable on day 2? $B_2 <$ € _____

(iii) As a fraction of the original €30: below which fraction of the money is a no longer affordable on day 2? _____ (decimal, two digits).

(k) **Burn the same fraction of the money on both days.** Multiply *both* budgets by the same number e (between 0 and 1): day 1 becomes $e \cdot 40$, day 2 becomes $e \cdot 30$.

(i) For $e = 0.9$, are Aman's choices rational?

[yes / no]

(ii) Below which value of e are Aman's choices rational? (At this value itself, the contradiction is still there.) $e =$ _____

(l) What is Andrey's e (Part B)? $e =$ _____

The number e has a name: the Critical Cost Efficiency Index (CCEI) (Afriat, 1972). Economists use it to measure *how rational* people are (Choi, Kariv, Müller and Silverman, 2014, "Who is (more) rational?"). Here is the idea, step by step.

1. Find the chains. A contradiction is a chain of "better than" links that comes back to where it started: choice 1 better than choice 2, and choice 2 better than choice 1 (or longer: 1 better than 2, 2 better than 3, 3 better than 1).

2. A chain breaks when one of its links breaks. The link "1 better than 2" exists because choice 2 was affordable on day 1. Burn enough of day 1's money that choice 2 no longer fits, and the link is gone.

3. We burn the same fraction of the money every day. So for each link we ask: what fraction of that day's money can we keep before the other choice no longer fits?

4. Break each chain at its cheapest link, the one that needs the least money burned: the link with the *largest* fraction (we must keep less than it, so we burn the least).

5. Every chain must be broken. e is the threshold: keep any fraction below e, and every chain is broken. So the least we must burn to remove every contradiction is *just over* $1 - e$ of the money.

Example (made-up numbers). Day 1: €50, and choice 2 costs €30 at day-1 prices. Day 2: €80, and choice 1 costs €76 at day-2 prices. One chain: 1 better than 2, 2 better than 1.

- Break "1 better than 2": day 1 must keep less than €30 of its €50, i.e. less than $\frac{30}{50} = 0.6$ of the money (burn more than 40 %).
- Break "2 better than 1": day 2 must keep less than €76 of its €80, i.e. less than $\frac{76}{80} = 0.95$ (burn more than 5 %).
- The cheapest break burns just over 5 %, so $e = 0.95$. If a second chain needed us to keep less than 0.65, we would have to burn more than 35 %, and e would be 0.65.

$e = 1$: nothing has to be burned, fully rational. The smaller e, the more money has to be burned, the further from rational.

Further reading (optional): There is a paper by James Andreoni and John Miller (2002), "Giving according to GARP: an experimental test of the consistency of preferences for altruism", *Econometrica* 70(2), 737–753 (free on Andreoni's website). Andreoni is an economist known for his work on giving and the "warm glow" of giving. In their experiment, people split money between themselves and someone else on many budget lines, and most of their choices pass the same test as in this problem; see pp. 739–743 (GARP on p. 739, the budgets on p. 740, the results and Afriat's index e on pp. 742–743).

Problem 3 · Whom to save

The story. A rescue team plans how to use its time in three emergencies. Let

$$x = \text{older people saved}, \qquad y = \text{children saved}.$$

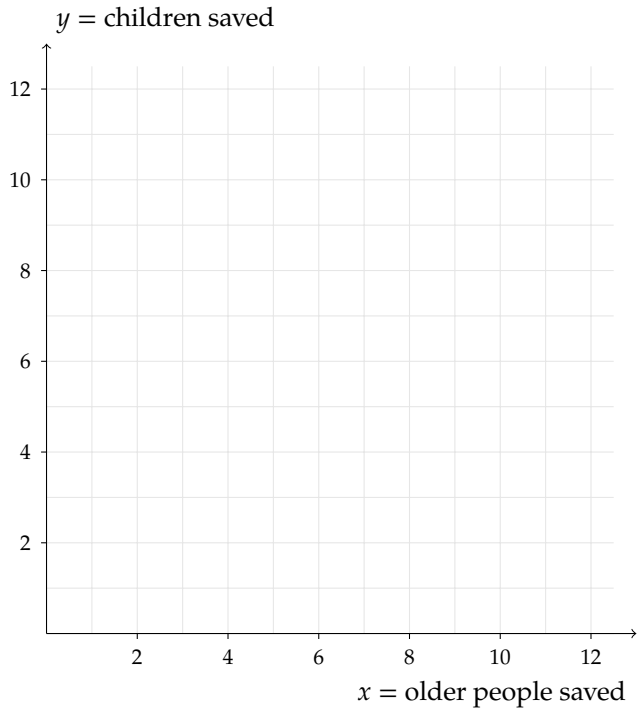
Saving a child always takes **10 minutes**. Saving an older person takes 5, 10 or 20 minutes, depending on the conditions. A plan (x, y) is **affordable** in a scenario if it takes at most the minutes available there. Minutes the team does not use are lost: nobody is saved in them.

- **Scenario 1.** The team has **60 minutes**; saving an older person takes 5 minutes: $5x + 10y = 60$.
- **Scenario 2.** The team has **60 minutes**; saving an older person takes 10 minutes: $10x + 10y = 60$.
- **Scenario 3.** The team has **120 minutes**; saving an older person takes 20 minutes: $20x + 10y = 120$.

Assumption. For the team, **more is better**: a plan that saves at least as many of both groups, and strictly more of one, is strictly better.

The team's plans. Scenario 1: plan 1 $= (2, 5)$. Scenario 2: plan 2 $= (2, 4)$. Scenario 3: plan 3 $= (5, 2)$.

(a) Draw the three budget lines in the graph, with their end points, and mark the three plans.



(b) Check that plan 3 uses all the time in scenario 3: $20 \cdot 5 + 10 \cdot 2 =$ _____ minutes.

(c) Scenario 1.

(i) How many minutes would plan 3 take in scenario 1? _____

(ii) Is plan 3 affordable in scenario 1?

[yes / no]

(iii) How many minutes would it leave unused? _____

(iv) Given that more is better for the team, what does its choice in scenario 1 tell us?

- ☐ plan 3 is strictly better than plan 1
- ☐ nothing
- ☐ plan 1 is strictly better than plan 3
- ☐ plan 1 is only weakly better than plan 3

(d) Scenario 3.

(i) How many minutes would plan 1 take in scenario 3? _____

(ii) Is plan 1 affordable in scenario 3?

[yes / no]

(iii) How many minutes would it leave unused? _____

(iv) Given that more is better for the team, what does its choice in scenario 3 tell us?

- ☐ plan 3 is strictly better than plan 1
- ☐ plan 3 is only weakly better than plan 1
- ☐ nothing
- ☐ plan 1 is strictly better than plan 3

(e) Scenario 2. Could the team have done plan 1 instead of plan 2? It takes $10 \cdot 2 + 10 \cdot 5 =$ _____ minutes. Affordable?

[yes / no]

(f) Using (c), (d) and (e): which pairs of plans contradict each other? Compare each pair using only its own two scenarios: in each of the two scenarios, was the other plan affordable? (A pair contradicts if each plan is revealed strictly better than the other, or one strictly and the other at least as good.)

- ☐ plans 1 and 2 only
- ☐ plans 1 and 3 only
- ☐ plans 2 and 3 only
- ☐ plans 1 and 2, and plans 1 and 3
- ☐ no pair

(g) How far is the team from rational? Shift all three budget lines inwards by the same fraction e (between 0 and 1), as if some time were burned: scenario 1 keeps $e \cdot 60$ minutes, scenario 2 keeps $e \cdot 60$, scenario 3 keeps $e \cdot 120$.

(i) For $e = 0.9$, could the three plans come from rational preferences for which more is better?

[yes / no]

(ii) What is the largest e at which they can? $e =$ _____ (decimal)

Further reading (*optional*)*:* There is a paper by Ricardo Andrés Guzmán, María Teresa Barbato, Daniel Sznycer and Leda Cosmides (2022), "A moral trade-off system produces intuitive judgments that are rational and coherent and strike a balance between conflicting moral values", *PNAS* 119(42), e2214005119 (open access). Cosmides is an evolutionary psychologist. They give people moral choices on 21 budget sets, as in this problem, and test them for rationality; see "Measuring Moral Rationality" and "Subjects Made Rational Judgments" in the results.

Problem 4 · Time for Anastassia and Adeeb

The story. Moriah has two kids, Anastassia and Adeeb, and some free hours each week to spend with them. Let

$$x = \text{hours with Anastassia}, \qquad y = \text{hours with Adeeb}.$$

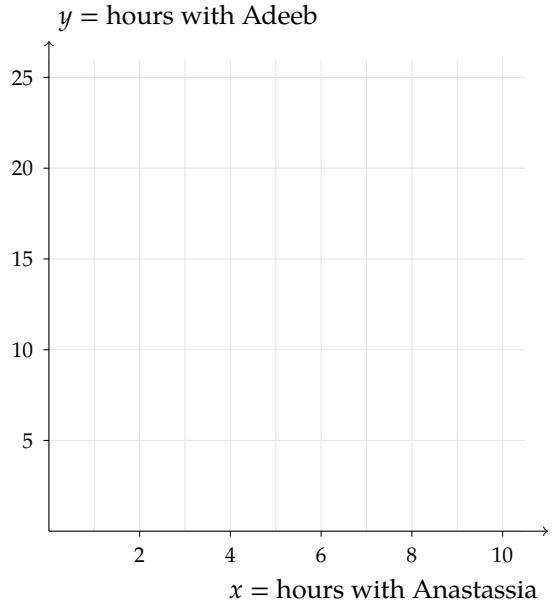
Adeeb still lives at home: an hour with Adeeb uses up one hour of Moriah's free time ($p_y = 1$). Anastassia studies in another city, so an hour with Anastassia also costs travel time: p_x is the number of free hours that one hour with Anastassia uses up. Free hours Moriah does not use are lost.

week		p_x	B (free hours)	budget line	Moriah's choice (x, y)
1	by train	3	20	$3x + y = 20$	$(6, 2)$
2	Anastassia is home for the holidays	1	10	$x + y = 10$	$(2, 8)$
3	train strike, slow bus	5	25	$5x + y = 25$	$(4, 5)$

Assumption. For Moriah, **more is better**: more time with one kid, and no less with the other, is strictly better.

Her friends say she secretly likes Anastassia more. Her choices won't settle that, but they can tell us whether they could come from rational preferences.

(a) Draw the three budget lines in the graph, with their end points, and mark the three choices.



(b) How many free hours would each choice take at each week's prices? Fill in the six empty cells (the diagonal is given).

	choice 1 $(6, 2)$	choice 2 $(2, 8)$	choice 3 $(4, 5)$
week 1 prices ($B = 20$ hours)	20	_____	_____
week 2 prices ($B = 10$ hours)	_____	10	_____
week 3 prices ($B = 25$ hours)	_____	_____	25

(c) Which pairs of weeks contradict each other? Compare each pair using only its own two weeks: in each of the two weeks, was the other week's choice affordable? (A pair contradicts if each choice is revealed strictly better than the other, or one strictly and the other at least as good.)

- ☐ weeks 1 and 2 only
- ☐ weeks 1 and 3 only
- ☐ weeks 2 and 3 only
- ☐ weeks 1 and 2, and weeks 2 and 3
- ☐ all three pairs

(d) **(Optional: no points, but its content can come up in the midterm and final.)** How far is Moriah from rational? Burn the same fraction of her free time every week, so that she keeps only the fraction e (between 0 and 1): week 1 keeps $e \cdot 20$ hours, week 2 keeps $e \cdot 10$, week 3 keeps $e \cdot 25$.

> **How to get** e (as in the box at the end of Problem 2). Find the chains (here the pairs of weeks from (c) decide e; the longer chain through all three weeks breaks earlier). A chain breaks once one of its links breaks: burn enough free time in one of its weeks that the other week's choice no longer fits. Break each chain at its cheapest link. e is the threshold: keep any fraction of her free time below e, and every chain is broken.

Below which e could Moriah's choices come from rational preferences for which more is better? $e =$ _____ (decimal)

Further reading (optional): There is a paper by Syngjoo Choi, Shachar Kariv, Wieland Müller and Dan Silverman (2014), "Who is (more) rational?", *American Economic Review* 104(6), 1518–1550 (free version: NBER Working Paper 16791). 1,182 Dutch adults made 25 choices each on budget lines, and their index e is linked to income, education and wealth; see the introduction, which explains e (Afriat's "Critical Cost Efficiency Index").

Problem 5 · Rational choice

What you need (and nothing more).

1. **Options and a menu.** A *menu* is a short, finite list of at least one option you can choose from, for example three desserts: cake, ice cream, fruit. We call options A, C, D.
2. **At least as good as.** $A \succsim C$ means: "A is at least as good as C" (you like A more than C, or equally).
3. **Strictly better.** $A \succ C$ means: "A is strictly better than C".
4. **Complete.** For any two options A and C: $A \succsim C$, or $C \succsim A$ (or both). You can always compare.
   Example: "is at least as tall as" (for any two people, one is at least as tall as the other).
   Not an example: "is taller than": two people of the same height, and neither is taller.
5. **Transitive.** If $A \succsim C$ and $C \succsim D$, then $A \succsim D$.
   Example: "is at least as tall as".
   Not an example: "won the match against": three teams can each beat one of the others in a circle.
6. **Rational choice.** From a menu, choose a *best* option: one that is at least as good as every option on the menu. In symbols, A is best on the menu if $A \succsim C$ for every option C on the menu.

Why this matters: does a best option always exist?

Best option. Recall item 6 of "What you need": A is a *best* option on a menu if $A \succsim C$ for every option C on the menu. In words: A is at least as good as everything on the menu. Ties are fine, so a menu can have several best options.

Example. Menu {tea, coffee, juice} with tea $\sim$ coffee $\succ$ juice. Tea and coffee are both best options: each is at least as good as everything on the menu. Juice is not. With ties there can be more than one best option, and any of them is a rational choice. That's why we say "*a* best option", not "*the* best option".

Why we care. "Choose a best option" (rational choice) only makes sense if a best option *exists*. Here are two three-option menus where it doesn't.

1. A cycle. Desserts: cake $\succ$ ice cream, ice cream $\succ$ fruit, and fruit $\succ$ cake (like rock–paper–scissors). Each dessert loses to another one: there is no best dessert.
What rules this out: transitivity. From cake $\succsim$ ice cream and ice cream $\succsim$ fruit, transitivity gives cake $\succsim$ fruit, so the circle can't close.

2. Can't compare. After your degree: a job in Vienna, a master's abroad, or a gap year. You rank the master's $\succsim$ the gap year, but you simply can't compare the job with either of them. Then no option is at least as good as all three, so there is no best option.
What rules this out: completeness. For any two options A and C, $A \succsim C$ or $C \succsim A$, so nothing is left uncompared.

So the claim we prove in Part A is exactly what is needed: if $\succsim$ is complete and transitive, the two failures above can't happen, and every menu has a best option.

Part A (worked for you): with $\succsim$, complete and transitive preferences always allow a rational choice.
Claim. If $\succsim$ is complete and transitive, then every menu has a best option.
Why, step by step, for a menu of three options A, C, D. First a small fact: completeness, used for

an option and itself, gives $A \gtrsim A$ (every option is at least as good as itself).

1. *Compare A and C.* By completeness, $A \gtrsim C$ or $C \gtrsim A$. **Without loss of generality**, say $A \gtrsim C$. (If instead $C \gtrsim A$, swap the names A and C: the rest of the argument is the same.)
2. *Compare A and D.* By completeness, $A \gtrsim D$ or $D \gtrsim A$. Two cases:
   - $A \gtrsim D$. Then $A \gtrsim A$, $A \gtrsim C$ and $A \gtrsim D$: A is a best option.
   - $D \gtrsim A$. From $D \gtrsim A$ and $A \gtrsim C$, transitivity gives $D \gtrsim C$. So $D \gtrsim A$, $D \gtrsim C$ and $D \gtrsim D$: D is a best option.
3. Either way the menu has a best option, and choosing it is a rational choice. ■

With more options it works the same way: keep the best option so far, compare it with the next one, and if the new one wins, transitivity passes all the old wins on to it.
Why both properties matter. Without completeness, step 1 could fail: "I can't compare them". Without transitivity, the second case of step 2 could fail: in a circle, like the three teams, every option is beaten by another, and there is no best option.

Part B (your turn): start from "strictly better" instead.

Now suppose we only know $\succ$ (strictly better), and two properties of it:

7. **Asymmetric.** If $A \succ C$, then *not* $C \succ A$. (Two options can't each be strictly better than the other.)
   Example: "is taller than". *Not an example:* "is at least as tall as" (two people of the same height).
8. **Negatively transitive.** If *not* $A \succ C$ and *not* $C \succ D$, then *not* $A \succ D$.
   Example: "is taller than": if Anna is not taller than Ben, and Ben is not taller than Carl, then Anna is not taller than Carl.
   Not an example: "is at least 5 cm taller than", for people of 170, 174 and 178 cm: 178 is not 5 cm taller than 174, 174 is not 5 cm taller than 170, yet 178 *is* 8 cm taller than 170.

Build "at least as good" from "strictly better":

$$A \gtrsim C \quad \text{means:} \quad \text{not } C \succ A \qquad (\text{"}C \text{ is not strictly better than } A\text{"}).$$

Reminder (see the box "Why this matters" before Part A): a best option is one that is at least as good as every option on the menu. *Show that this $\gtrsim$ is complete and transitive.* Then Part A applies, and every menu has a best option. (In the proof you will use definition 8 with the options in a different order: D in place of A, and A in place of D.)

(B1) **This $\gtrsim$ is complete.** Take any two options A and C. Suppose, to the contrary, that **neither** $A \gtrsim C$ **nor** $C \gtrsim A$.

(i) "not $A \gtrsim C$" means, by the definition:

- ☐ not $C \succ A$
- ☐ $A \succ C$
- ☐ $C \succ A$

(ii) "not $C \gtrsim A$" means, by the definition:

- ☐ $A \succ C$
- ☐ $C \succ A$
- ☐ not $A \succ C$

(iii) $C \succ A$ and $A \succ C$ together are impossible. Which property of $\succ$ says so?

- ☐ negatively transitive
- ☐ complete
- ☐ asymmetric

So $A \succsim C$ or $C \succsim A$: this $\succsim$ is complete.

(B2) **This $\succsim$ is transitive.** Suppose $A \succsim C$ and $C \succsim D$.

(i) By the definition, $A \succsim C$ means

- ☐ $C \succ A$
- ☐ not $A \succ C$
- ☐ not $C \succ A$

and $C \succsim D$ means

- ☐ not $C \succ D$
- ☐ not $D \succ C$
- ☐ $D \succ C$

(ii) Use negative transitivity (definition 8) with D, C, A in place of A, C, D: "not $D \succ C$ and not $C \succ A$" gives

- ☐ $D \succ A$
- ☐ not $D \succ A$
- ☐ not $A \succ D$

(iii) By the definition, "not $D \succ A$" means

- ☐ $A \succsim D$
- ☐ $A \succ D$
- ☐ $D \succsim A$

So $A \succsim D$: this $\succsim$ is transitive.

(B3) **The conclusion.** So this $\succsim$ is complete and transitive, and by Part A every menu has a best option (best = at least as good as every option on the menu; see the box "Why this matters" before Part A). In words, with this $\succsim$, a best option on a menu is one that

- ☐ no option on the menu is strictly better than
- ☐ is at least as good as at least one option on the menu
- ☐ is strictly better than every other option

Part C · Quick checks.

Two more properties. For a relation R between options (like "is taller than"):
Symmetric: if $A\ R\ C$, then $C\ R\ A$.
Antisymmetric: if $A\ R\ C$ and $C\ R\ A$, then A and C are the same option.

(C1) Which properties does "is the same height as" (between people) have?

1. complete
2. transitive
3. symmetric

- ☐ 1 only
- ☐ 3 only
- ☐ 1 and 2
- ☐ 2 and 3
- ☐ 1, 2 and 3

(C2) Is "is a friend of" transitive?

[yes / no]

(C3) Is "is at least as tall as" antisymmetric?

[yes / no]

(C4) You have complete and transitive preferences $\succsim$, and you like cake and fruit equally (cake $\sim$ fruit). Which statements are correct?

1. $\sim$ is symmetric.
2. $\succ$ is asymmetric.
3. Your $\succsim$ is antisymmetric.

- ☐ 1 only
- ☐ 2 only
- ☐ 1 and 2
- ☐ 2 and 3
- ☐ 1, 2 and 3

We will come back to these concepts later in the course: Condorcet's paradox, Borda voting and Arrow's impossibility theorem.

Further reading (optional): There is a paper by Amos Tversky (1969), "Intransitivity of preferences", *Psychological Review* 76(1), 31–48 (DOI 10.1037/h0026750, through the university library). Tversky later built prospect theory with Daniel Kahneman. In his experiments some people choose in a circle, as in the dessert example; see the experiments and their results. For the definitions: Frank, 9th ed., ch. 3, pp. 63–64 (completeness and transitivity).

Solutions

Problem 1

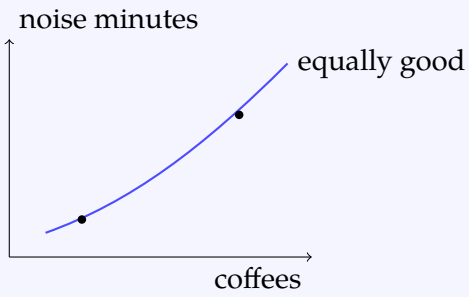
(a) 2 only. Complete means: for any two options you can say which you like more, *or* that you like them equally. Statement 1 ("exactly the same") *is* a comparison: indifference. Statement 3 ("strictly better") is a clear comparison. Only statement 2 ("I can't compare them at all") has no answer, so it breaks completeness.
Typical mistake: "1 and 2", reading "I like them the same" as "I can't decide". Indifference is a decision: both are fine.

(b) $(5, 30)$. More is better only ranks a bundle that has at least as much of *both* goods and strictly more of one. $(5, 30)$: one more ticket, the same euros: better, by more is better. $(5, 25)$ and $(3, 40)$: more of one good, less of the other; this property alone can't say, it depends on the person's preferences. $(3, 29)$: less of both, so worse. $(4, 30)$ is A itself: as good, not better.

(c) 1 and 2. Statement 1: on one curve all bundles are equally good, so $C \sim A$ (curve 1) and $A \sim D$ (curve 2): true. Statement 2: equally good as A, and A equally good as D; by transitivity $C \sim D$: true. Statement 3: D has more of both goods than C, so by more is better $D \succ C$. "Equally good" and "strictly better" cannot both hold, so this drawing cannot come from preferences that are transitive and satisfy more is better: false. That is why indifference curves never cross.
Why the weak version of more is better is enough: D has more of both goods than C. Even the weak version (strictly more of every good is strictly better) then says $D \succ C$, which contradicts $C \sim D$. The strong version is not needed here.
Which properties rule out this drawing? **Transitivity and more is better.** Transitivity alone allows it (if D were not better than C, "$C \sim A \sim D$" would be fine), and more is better alone allows it (without transitivity, $C \sim A$ and $A \sim D$ need not give $C \sim D$). Only both together give $C \sim D$ *and* $D \succ C$, a contradiction. Completeness plays no role here. "None" is wrong: with both properties the drawing is impossible, as statement 3 shows.

(d) (i) **more of both goods**: $6 > 2$ tickets and $30 > 10$ euros. (ii) **strictly better than** A, by more is better. (iii) **equally good**: that is what lying on one indifference curve means. (iv) **more is better.** (ii) and (iii) contradict each other: C cannot be strictly better than A and equally good at the same time. When both goods are goods, moving up and to the right always reaches a better bundle, so an indifference curve must slope *down*: to stay equally well off, more of one good has to be paid for with less of the other.
$C = (6, 30)$ has more of both goods than $A = (2, 10)$, so already the weak version of more is better says $C \succ A$. That contradicts "equally good". If one of the goods were a bad (like minutes of noise), an upward-sloping curve would be perfectly possible.

When a "good" is a bad. Noise minutes are a *bad*: less is better. Local non-satiation still holds, but now an indifference curve slopes *up*: to stay equally happy with more noise, you need more coffee.

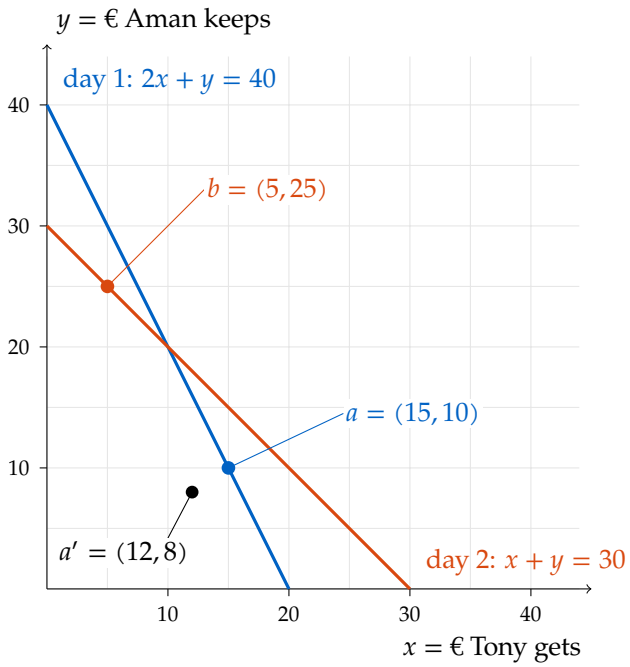


So local non-satiation *alone* cannot rule out an upward-sloping indifference curve: it says something better is nearby, but not in which direction. For the curve questions we need "more is better" (the weak version is enough).

Which question needs which?

- *Comparing options* (completeness): none of the three.
- *"Using only more is better, which of these bundles is better?"* (more of one good, the same of the other): the **strong** version. With the weak version only, such a bundle would not have to be better. With local non-satiation only, we know that something better is nearby, but not which bundle.
- *The upward-sloping curve* and *the crossing curves*: the **weak** version is enough. In both, one bundle has more of *both* goods than the other; the weak version already says that bundle is strictly better, and that contradicts "equally good". Local non-satiation alone is not enough: it does not say in which direction things get better, and here the direction is the whole point.
- *Keep it or give it*, "money left over, so the chosen split is strictly better" (Aman, Andrey, and how much money to take away): **local non-satiation on its own is enough**. Euros kept and euros given are both goods, not bads, so local non-satiation clearly holds there. And it is all this step needs, because no direction is needed: with money left over, *every* split close to the cheaper one was affordable, one of them is better than it, and the chosen split was picked over that one too. (This step would work even if Aman disliked giving; for the curve questions, by contrast, we need to know which direction is better.)

Problem 2 · Keep it or give it



(a) Day 1: $2x + y = 40$, from $(0, 40)$ (Aman keeps everything) to $(20, 0)$ (he sends all €40, and Tony gets half, €20). Day 2: $x + y = 30$, from $(0, 30)$ to $(30, 0)$. The two lines cross at $(10, 20)$.

(b) More is better. $a = (15, 10)$ gives Tony more $(15 > 12)$ *and* gives Aman more $(10 > 8)$, so a is strictly better. (Euros for Aman and euros for Tony are both goods, so more of both is better: the weak version of more is better is already enough here.)

(c) Completeness. Without it, "which one does he like more?" might have no answer, and a choice would tell us nothing.

(d) (i) a' costs $2 \cdot 12 + 8 = 24 + 8 = 32$ at day-1 prices, less than €40: **yes**. On the graph, a' lies below the day-1 line.
(ii) $40 - 32 =$ **€8** would be burned.
(iii) a **is strictly better than** a'**.** Aman could afford a' and chose a, so $a \succsim a'$. Since a' burns €8 and a uses all the money, more is better makes it strict: $a \succ a'$.
Here local non-satiation on its own is enough. Euros for Aman and euros for Tony are both goods, so near any split there is a better one. No direction is needed: with the €8 left over, every split close to a' was affordable, one of them is better than a', and Aman chose a over it.

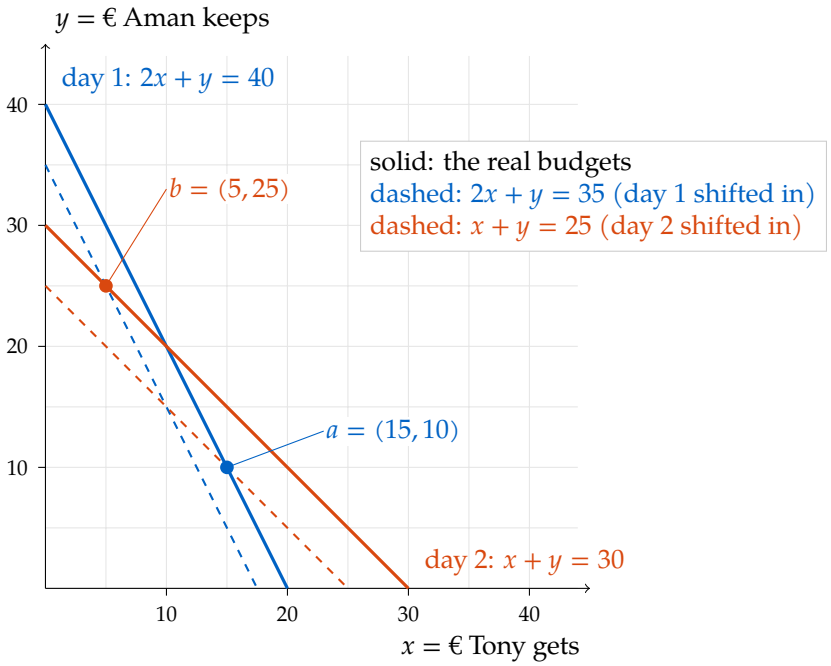
(e) (i) At day-2 prices a costs $15 + 10 = 25$, less than €30: **yes**. On the graph, a lies below the day-2 line.
(ii) $30 - 25 =$ **€5** would be burned.
(iii) b **is strictly better than** a**.** He could afford a and chose b; a burns €5 and b uses all €30: $b \succ a$.

(**f**) At day-1 prices b costs $2 \cdot 5 + 25 = 10 + 25 = 35$, less than €40: **yes**, with **€5** burned. On the graph, b lies below the day-1 line. He could afford b on day 1 and chose a, which uses all €40: a **is strictly better than** b, $a \succ b$.

(**g**) **No.** Day 1 says $a \succ b$; day 2 says $b \succ a$. No preferences that are complete and transitive, *and* for which more is better, can produce both choices. The choices form a cycle, $a \succ b \succ a$, so the property that breaks is **transitivity** (the same cycle as the money pump in L06): by transitivity Aman would like a strictly more than a itself.
Why transitivity is the culprit, step by step. The strict preference itself is built with transitivity. On day 1, Aman could have had b *plus* the €5 he would burn, say $b^+ = (5, 30)$, which costs exactly €40. He chose a, so $a \succsim b^+$. More is better gives $b^+ \succ b$. Transitivity then gives $a \succ b$. The same steps on day 2, with $a^+ = (15, 15)$, give $b \succ a$. So the test checks completeness, transitivity and more is better *together*; given that more is better, transitivity is what breaks.
The pattern to remember: Aman gave *more* when giving was *dearer* (€15 to Tony at a price of 2, only €5 at a price of 1). With two goods, every such contradiction between two choices is a move in the wrong direction like this one.
Without any form of more is better, Aman's choices are rational: yes. Then his choices only reveal *weak* preferences, $a \succsim b$ (day 1) and $b \succsim a$ (day 2), which together say he likes a and b *equally*, $a \sim b$. That is complete and transitive: nothing is violated. It is the assumption that more is better (no one wants to burn money), that turns both into *strict* preferences and creates the contradiction. So the test checks rationality *together with* more is better.
Careful: even the weakest form of more is better (close to any bundle there is always a slightly better one) makes "money left unused" reveal a *strict* preference: with the €5 Aman would burn, something slightly better than b was affordable on day 1, and he chose a over that too. So the contradiction comes back. The choices are rational only if we assume nothing at all about more being better.

(**h**) **Yes.** Day 1: his day-2 split $(8, 22)$ costs $2 \cdot 8 + 22 = 38 \leq 40$, so it was affordable (€2 burned) and he chose $(5, 30)$: $(5, 30) \succ (8, 22)$. Day 2: his day-1 split $(5, 30)$ costs $5 + 30 = 35 > 30$, not affordable, so day 2 tells us nothing about these two. Only one "strictly better", no cycle: his altruism is **rational**. (He gives a little more when giving gets cheaper.)

Part C



(i) (i) b costs $2 \cdot 5 + 25 = 35$ at day-1 prices: **€35**.
(ii) b is no longer affordable once $B_1 <$ **€35**: the dashed blue line $2x + y = 35$ passes exactly through b; any line below it leaves b outside.
(iii) **No.** Day 1 only tells us about options Aman *could* have chosen.
(iv) **Yes.** Only day 2's statement ($b \succ a$) is left, so there is no cycle.
(v) Below $35/40 =$ **0.875**: we need to take away just over 12.5 % of the day-1 money.

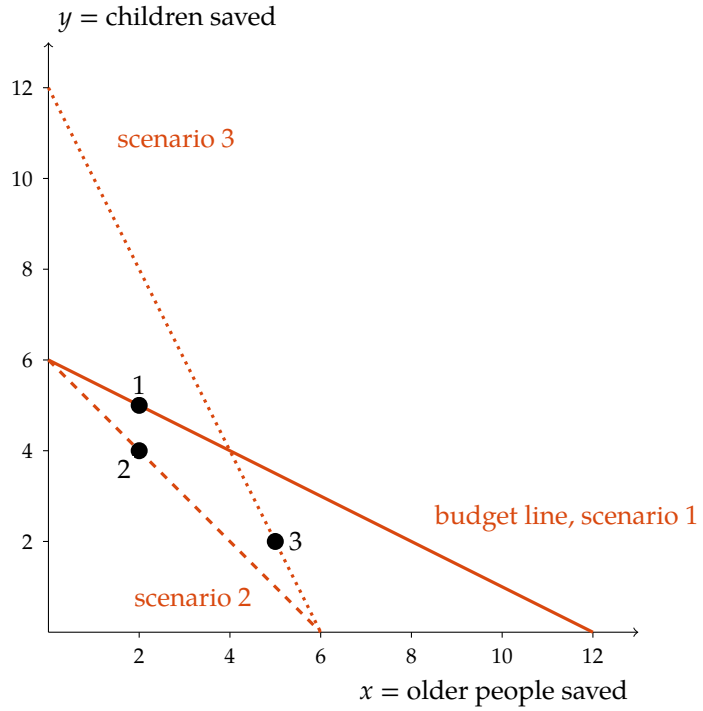
(j) (i) a costs $15 + 10 = 25$ at day-2 prices: **€25**.
(ii) $B_2 <$ **€25** (dashed orange line $x + y = 25$ through a).
(iii) Below $25/30 \approx$ **0.83**: just over 16.7 % of the day-2 money has to go. Shifting day 2 alone costs more than shifting day 1 alone.

(k) (i) $e = 0.9$: day 1 has $0.9 \cdot 40 =$ €36 ≥ 35, so b is still affordable and $a \succ b$; day 2 has $0.9 \cdot 30 =$ €27 ≥ 25, so a is still affordable and $b \succ a$. The cycle is still there: **no**.
(ii) To break the cycle it is enough to break *one* of the two statements. Day 1 breaks when $e \cdot 40 < 35$, i.e. $e < 0.875$; day 2 breaks when $e \cdot 30 < 25$, i.e. $e < 0.83$. The easier one decides: for every e below **0.875** the choices are rational, at 0.875 and above they are not. So $e =$ **0.875**.

(l) Andrey's choices are rational with the full budgets, so nothing has to be taken away: $e = 1$.

What you have just computed. $e = 0.875$ is Aman's **CCEI** (the index goes back to Afriat, 1972): his choices are rational once we accept that just over 12.5 % of his money is "wasted". In the lab, the CCEI is computed exactly like this, only with many more budget lines (e.g. 25–50 per person): shift all budget lines inwards by the same fraction (burn budget) until no cycle is left.

Problem 3



plan 1 = (2, 5) plan 2 = (2, 4) plan 3 = (5, 2)

(a) Scenario 1 runs from (0, 6) (only children) to (12, 0) (only older people); scenario 2 from (0, 6) to (6, 0); scenario 3 from (0, 12) to (6, 0). Every plan lies on its own line: $5 \cdot 2 + 10 \cdot 5 = 60$; $10 \cdot 2 + 10 \cdot 4 = 60$; $20 \cdot 5 + 10 \cdot 2 = 120$.

(b) $100 + 20 = 120$: exactly the 120 minutes. **120.**

(c) (i) Price plan 3 with scenario-1 times: $5 \cdot 5 + 10 \cdot 2 = 25 + 20 = 45$. **45.**
(ii) 45 is less than 60: **yes**.
(iii) $60 - 45 =$ **15** minutes unused.
(iv) With 15 minutes left the team could have saved more people than in plan 3; by more is better that is strictly better than plan 3, and the team chose plan 1 over it. So **plan 1 is strictly better than plan 3**.
Typical mistake: pricing plan 3 with scenario-3 times (120). Always use the times of the scenario you are checking (here scenario 1, where plan 1 was chosen).

(d) (i) $20 \cdot 2 + 10 \cdot 5 = 40 + 50 = 90$. **90.**
(ii) 90 is less than 120: **yes**.
(iii) **30** minutes unused.
(iv) Same argument: **plan 3 is strictly better than plan 1**.

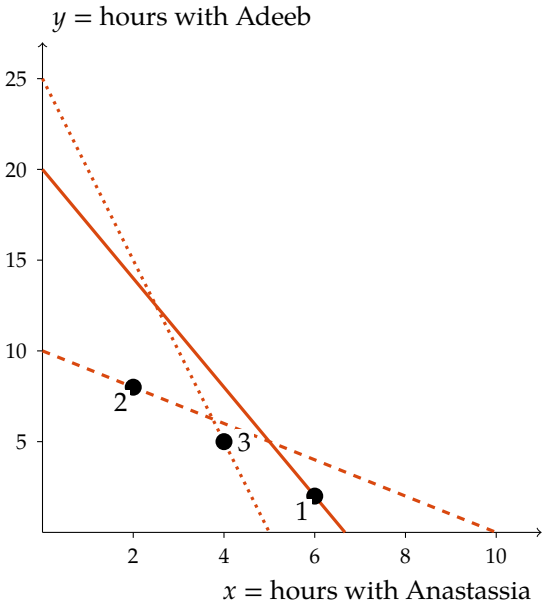
(e) $20 + 50 = 70$ minutes, more than 60: **70; no**. Plan 3 in scenario 2: $50 + 20 = 70$ too, not affordable.
All clues together (every cross time): scenario 1: plan 2 takes 50, plan 3 takes 45 (both affordable); scenario 2: plans 1 and 3 take 70 each (neither affordable); scenario 3: plan 1 takes 90, plan 2 takes 80 (both affordable).

(f) Plans 1 and 3 only.
Plans 1 and 3: plan 3 takes $45 < 60$ minutes in scenario 1 (plan 1 $\succ$ plan 3), and plan 1 takes $90 < 120$ in scenario 3 (plan 3 $\succ$ plan 1): a contradiction.
Plans 1 and 2: plan 2 takes $50 < 60$ in scenario 1 (plan 1 $\succ$ plan 2), but plan 1 takes $70 > 60$ in scenario 2, so it was not affordable there. One direction only: no contradiction.
Plans 2 and 3: plan 2 takes $80 < 120$ in scenario 3 (plan 3 $\succ$ plan 2), but plan 3 takes $70 > 60$ in scenario 2. One direction only: no contradiction.
In words: when older people were quick to save (5 minutes) the team saved only 2 of them; when they were slow (20 minutes) it saved 5.

(g) (i) $e = 0.9$: scenario 1 has 54 minutes and plan 3 takes 45 (affordable, 9 left): plan 1 $\succ$ plan 3. Scenario 3 has 108 minutes and plan 1 takes 90 (affordable, 18 left): plan 3 $\succ$ plan 1. Still a contradiction: **no**.
(ii) Plan 3 takes 45 of scenario 1's 60 minutes, a fraction $\frac{45}{60} = \frac{3}{4}$ of the time. Plan 1 takes 90 of scenario 3's 120, also $\frac{90}{120} = \frac{3}{4}$. At $e = 0.75$ scenario 1 has 45 minutes and scenario 3 has 90: each plan takes *exactly* the time left in the other scenario after the shift. Then each choice says only "at least as good", a tie, and two ties do not contradict each other. (Plan 2: in scenario 1 it now needs $50 > 45$, not affordable; in scenario 3 it needs $80 < 90$, so plan 3 $\succ$ plan 2; no contradiction.) Above 0.75 both comparisons are strict again. **0.75.**
The number e is the CCEI from Problem 2: here the plans become rational once we treat 25 % of the time as wasted.

Why this matters. Researchers test moral choices the same way: Guzmán, Barbato, Sznycer and Cosmides (2022, *PNAS*) gave 1,745 people 21 budget sets in each of two framings, for sparing civilians against saving soldiers in a war dilemma; 71–78 % never made choices that contradicted rational preferences. Rational is not the same as moral: a team that never saves older people can be perfectly rational.

Problem 4



budget lines: week 1 solid, week 2 dashed, week 3 dotted

choice 1 = $(6, 2)$ choice 2 = $(2, 8)$ choice 3 = $(4, 5)$

(a) Week 1 runs from $(0, 20)$ to $(\frac{20}{3}, 0)$; week 2 from $(0, 10)$ to $(10, 0)$; week 3 from $(0, 25)$ to $(5, 0)$. Every choice lies on its own line: $3 \cdot 6 + 2 = 20$; $2 + 8 = 10$; $5 \cdot 4 + 5 = 25$.

(b) Always use the prices (hours per hour) of the row's week.
Week 1 $(3x + y)$: choice 2 takes $6 + 8 =$ **14** hours; choice 3 takes $12 + 5 =$ **17** hours.
Week 2 $(x + y)$: choice 1 takes $6 + 2 =$ **8** hours; choice 3 takes $4 + 5 =$ **9** hours.
Week 3 $(5x + y)$: choice 1 takes $30 + 2 =$ **32** hours; choice 2 takes $10 + 8 =$ **18** hours.

(c) Weeks 1 and 2, and weeks 2 and 3.
Weeks 1 and 2: $14 < 20$ (choice 1 $\succ$ choice 2) and $8 < 10$ (choice 2 $\succ$ choice 1): a contradiction.
Weeks 2 and 3: $9 < 10$ (choice 2 $\succ$ choice 3) and $18 < 25$ (choice 3 $\succ$ choice 2): a contradiction.
Weeks 1 and 3: $17 < 20$ (choice 1 $\succ$ choice 3), but $32 > 25$: choice 1 was not affordable in week 3. One direction only: no contradiction between these two weeks.
Note: through the chain choice 3 $\succ$ choice 2 $\succ$ choice 1, week 3 is revealed better than week 1 indirectly. That is why the question says "using only its own two weeks". That longer chain breaks once she keeps less than 0.85 (its cheapest link, "1 better than 3", $\frac{17}{20}$), so it never decides e.

(d) (optional) **0.8.** Step by step, in chains and links:
(i) Chain weeks 1 and 2 (1 better than 2, and 2 better than 1). Link "1 better than 2" exists because choice 2 (14 hours) fits into week 1's 20 hours; it breaks once week 1 keeps less than 14 of its 20 hours, i.e. less than $\frac{14}{20} = 0.7$ of them. Link "2 better than 1" breaks once week 2 keeps less than 8 of its 10 hours, i.e. less than $\frac{8}{10} = 0.8$. The cheapest link is the one that needs the least burned: keep less than 0.8. So this chain is gone below **0.8**.
(ii) Chain weeks 2 and 3: link "2 better than 3" breaks below $\frac{9}{10} = 0.9$, link "3 better than 2" below $\frac{18}{25} = 0.72$. Cheapest link: keep less than 0.9. Gone below **0.9**.
(iii) Every chain must be broken, so she must keep less than 0.8 *and* less than 0.9: $e =$ **0.8**.

(The longer chain $1 \to 3 \to 2 \to 1$ has links that break below 0.85, 0.72 and 0.8, so it is already broken below 0.8.) At $e = 0.8$ itself, weeks 1 and 2 still contradict: week 2 keeps exactly 8 hours and choice 1 takes exactly 8 hours there, so choice 2 still counts as "at least as good" as choice 1; in week 1, 16 hours are left and choice 2 takes only 14, so choice 1 is still strictly better than choice 2.
Typical mistakes: 0.7 (breaking the chain at its expensive link: that burns more than needed); 0.9 (breaking only the second chain); 0.72.
The number e is the CCEI from Problem 2: Moriah's choices become rational once we treat more than 20 % of her free time as lost (e below 0.8).

Problem 5

(B1) (i) $C \succ A$. $A \succsim C$ means "not $C \succ A$"; so "not $A \succsim C$" is "not (not $C \succ A$)", which is $C \succ A$.
(ii) $A \succ C$: the same with A and C swapped.
(iii) **Asymmetric**: if $A \succ C$, then not $C \succ A$, so both cannot hold. So at least one of $A \succsim C$, $C \succsim A$ holds: this $\succsim$ is complete.
Typical mistakes: in (i), "not $C \succ A$" (dropping the outer "not") or "$A \succ C$" (swapping the letters); in (iii), "negatively transitive" (it needs three options and "not" statements) or "complete" (that is what we are showing about $\succsim$, not something we know about $\succ$).

(B2) (i) **not** $C \succ A$; **not** $D \succ C$: the definition, letter by letter ($A \succsim C$: "C is not strictly better than A").
(ii) **not** $D \succ A$: definition 8 with D first, C in the middle, A last: not $D \succ C$ and not $C \succ A$ give not $D \succ A$.
(iii) $A \succsim D$: "D is not strictly better than A" is exactly $A \succsim D$. So this $\succsim$ is transitive.
Typical mistakes: in (i), "not $A \succ C$" (reading the definition the wrong way round); in (ii), "not $A \succ D$" (that would be $D \succsim A$, a different statement); in (iii), "$D \succsim A$" (reversed) or "$A \succ D$" (not strictly better is not the same as strictly better).

(B3) **No option on the menu is strictly better than it.** Part A turns "complete and transitive" into "a best option exists". With $A \succsim C$ meaning "not $C \succ A$", "$A \succsim$ every option" means "no option is strictly better than A". Ties are allowed, so there can be more than one best option; "strictly better than every other option" is therefore wrong. "At least as good as at least one option" is too weak: every option is at least as good as itself, so every option would count as best.

Two checks behind the result.
We get the same "strictly better" back. With the new $\succsim$, "$A \succsim C$ and not $C \succsim A$" means "not $C \succ A$, and $A \succ C$", which is just $A \succ C$ (by asymmetry, $A \succ C$ already rules out $C \succ A$).
Why we need negative transitivity, not just transitivity. "Has strictly more of both goods" is asymmetric and transitive, but the $\succsim$ built from it is not transitive: with $A = (1,1)$, $C = (3,0)$, $D = (2,2)$ we get $A \succsim C$ (C does not have more of both than A) and $C \succsim D$ (D does not have more of both than C), yet D has more of both than A, so not $A \succsim D$.

(C1) **2 and 3.** Not complete: for two people of different heights, neither is "the same height as" the other. Transitive: if A is the same height as C, and C the same height as D, then A is the same height as D. Symmetric: if A is the same height as C, then C is the same height as A.

(C2) **No.** A friend of your friend need not be your friend.

(C3) **No.** Two different people of the same height are each at least as tall as the other, but they are not the same person.

(C4) **1 and 2.** 1: if $A \sim C$, then $C \sim A$. 2: $A \succ C$ and $C \succ A$ cannot both hold. 3 fails: cake $\succsim$ fruit and fruit $\succsim$ cake, but cake and fruit are different options. Ties rule out antisymmetry.